

SQL Query Result Explanation

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The SQL Query:

```
SELECT * FROM Employees  
WHERE FIRST_NAME BETWEEN 'a' AND 'c';
```

The Employees Table:

EmployeeID	FIRST_NAME	LAST_NAME	DEPARTMENT	SALARY
1	Ahmed	Ali	IT	5000
2	Bassem	Hassan	HR	4500
3	carlos	Mona	Finance	7000
4	David	Samy	IT	8000
5	Eman	Khaled	Sales	3500
6	Farah	Nabil	HR	4000
7	George	Adel	IT	9000
8	Hany	Maher	Finance	6000
9	Ibrahim	Mostafa	Sales	3000
10	John	Doe	IT	7500
11	Karim	Fathy	HR	4200
12	Laila	Saeed	Finance	5200
13	Mariam	Yasser	IT	6100
14	Nader	Ramy	Sales	3300
15	Omar	Tarek	HR	4700
16	Peter	Nagy	Finance	5800
17	Queen	Ali	IT	8200
18	Rana	Mahmoud	Sales	3900
19	Salma	Hesham	HR	4100
20	Tamer	Ibrahim	IT	6700

The Result:

ID	FIRST_NAME	LAST_NAME	DEPARTMENT	SALARY
1	Ahmed	Ali	IT	5000
2	Bassem	Hassan	HR	4500

The explanation of the result:

In SQL, when using the BETWEEN operator with strings, the engine compares characters in alphabetical order. The range 'a' to 'c' includes everything that starts with 'a' or 'b', but it stops exactly at the string 'c'.

Why 'Carlos' is NOT included:

The word 'Carlos' is alphabetically greater than 'c'. Because 'Carlos' has additional characters ('a', 'r', 'l', 'o', 's') following the initial 'C', it would appear after 'c' in a dictionary. Therefore, it falls outside the requested range.

Row-by-Row Analysis:

Ahmed: Starts with 'A' – Included (Between 'a' and 'c')

Bassem: Starts with 'B' – Included (Between 'a' and 'c')

Carlos: Alphabetically > 'c' – Excluded

David: Starts with 'D' – Excluded (After 'c')

Thanks